Continuous Sentence-Level Translation of Indian Sign Language to English

Case Study Report

TEAM-10

Project Overview

Problem Statement

India has an estimated 63 million people who are Deaf or Hard of Hearing (WHO, 2016), for whom Indian Sign Language (ISL) is a primary mode of communication. ISL has its own grammar, word order and non-manual markers (facial expression, mouth shape, body posture); it is not a signed encoding of English. This creates a persistent communication barrier between ISL users and the hearing, non-signing majority in education, healthcare, e-governance and everyday life.

The project addresses **continuous, sentence-level Sign Language Translation (SLT)**: given a video of a person signing a full sentence in ISL, the system must produce the English sentence with the same meaning, without first segmenting the video into isolated signs. This is harder than isolated-sign recognition because sign boundaries are not marked, ISL and English differ in word order and grammar, and meaning is carried jointly by the hands, body and face.

Motivation

- **Accessibility:** ISL→English translation can reduce dependence on human interpreters, who are scarce in India, for everyday services, classrooms and public institutions.
- **Research gap:** compared with American or German Sign Language, ISL has few annotated resources and benchmarked tasks, so there are far fewer data-driven translation systems.
- **Technical challenge:** SLT is a multimodal sequence-to-sequence problem that must fuse video with skeletal motion and map a long, unsegmented input to a short, fluent sentence.

Project Goal

Build a deep-learning model that translates recorded ISL sentence videos into English, evaluate it rigorously, and serve it through an API that the team's web application calls with webcam recordings.

1. Literature Survey

Five recent papers (2025), each in a journal indexed on Scimago Journal Rank, were reviewed:

1.1 Said et al. (2025) — Adaptive Transformer-Based Deep Learning Framework for Continuous Sign Language Recognition and Translation

Mathematics (MDPI), 13(6), 909, DOI 10.3390/math13060909. Proposes an Adaptive Transformer with adaptive masking, local clip self-attention and adaptive fusion to handle the weak alignment between long sign videos and short sentences. **Relevance:** informs the fusion of video and pose features before the decoder.

1.2 Arib, Akter, Rahman & Rahman (2025) — SignFormer-GCN: Continuous Sign Language Translation Using Spatio-Temporal Graph Convolutional Networks

PLOS ONE, DOI 10.1371/journal.pone.0316298. Combines a transformer with a spatio-temporal graph network over skeleton keypoints, arguing that RGB-only pipelines miss the structure of sign articulation. **Relevance:** supports keeping a dedicated pose branch — confirmed by this work's ablation test, where removing pose collapses accuracy.

1.3 Nedungadi, Dileep, Geetha & Raman (2025) — Vision Transformer-Powered Conversational Agent for Real-Time Indian Sign Language E-Governance Accessibility

Scientific Reports, 15, 33055, DOI 10.1038/s41598-025-12667-3. Deploys a CNN–transformer for continuous ISL recognition feeding a BART/T5-style engine on India's UMANG platform (97.5% accuracy, 6.53 s mean latency). **Relevance:** confirms BART-family decoders for ISL-to-text and provides the latency reference used in testing.

1.4 Pandey, Tahseen, Pathak, Parveen & Maurya (2025) — Real-Time Vision-Based Indian Sign Language Translation Using Deep Learning Techniques

IJIRCST, 13(3), DOI 10.55524/ijircst.2025.13.3.6. Compares CNN, LSTM and Transformer encoder–decoders on ISL/ASL data (BLEU 0.74, 96.1% accuracy) and uses BLEU, ROUGE, CER and WER. **Relevance:** informs the choice of metrics.

1.5 Gonfa, Meshesha, Girmaw & W/Maryam (2025) — A Deep Learning Framework for Ethiopian Sign Language Recognition Using Skeleton-Based Representation

Scientific Reports, 15, 36181, DOI 10.1038/s41598-025-19937-0. Uses MediaPipe skeletal keypoints with recurrent models and reports a large signer-dependent (94%) vs signer-independent (73%) gap. **Relevance:** mirrors this work's MediaPipe pose pipeline and its generalisation concerns.

The adversarial component follows the gradient-reversal method of Ganin & Lempitsky (2015), and the test strategy follows the CDAO framework *Test and Evaluation of Artificial Intelligence Models* (2024).

Member 1

Scope of this part: a dual-stream (pose + video) translation model with a pretrained BART decoder, fine-tuned on the ISL-CSLTR sentence corpus, an adversarial (GAN-style) modality-alignment experiment, and a structured test and evaluation of the final model.

2. Architecture Diagram for the Overall Application

The top row shows how the model is built: features are extracted from the ISL-CSLTR videos, the model is initialized from an iSign-pretrained checkpoint, fine-tuned with and without the GAN component, tested, and published to a private Hugging Face repository. The bottom row shows the application: the web app records the user signing, a Hugging Face Space runs the same feature extraction and the model, and returns the closest of the 101 known sentences together with a confidence score.

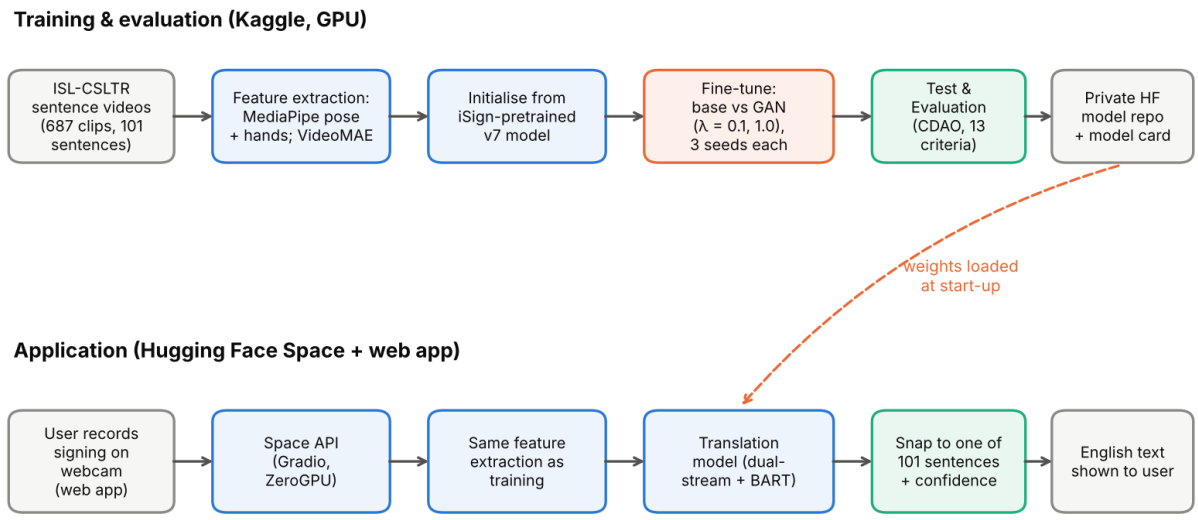


Figure 1: Overall application architecture.

3. Module Details

Module	Function	Implementation
Data preparation	Builds labels and train/validation/test splits	Sentence = video folder name; per sentence 1 test, 1 validation, rest training (687 videos, 101 sentences)
Pose extraction	Body and hand keypoints per frame	MediaPipe Tasks (pose + hand landmarkers); 75 points (33 body, 21 per hand) with x, y, z and confidence; centred on the shoulders and scaled by shoulder–nose distance
Video features	Visual summary of the clip	Frozen VideoMAE-base on 16 frames, pooled to 8 tokens of 768 values

Module	Function	Implementation
Skeleton encoder	Learns motion patterns over time	3-layer Transformer (256-dim, 4 heads) over up to 450 frames
Fusion	Combines both streams	Video and pose tokens joined into one sequence and projected to 768 dimensions
Decoder	Writes the English sentence	Pretrained BART-base; only its cross-attention layers are trained; beam search with 4 beams
Modality discriminator (GAN)	Aligns video and pose representations	Small network that tries to tell video from pose; a gradient-reversal layer trains the encoders to fool it
Sentence snapping	Maps output to a valid sentence	Picks the closest of the 101 known sentences (ROUGE-L similarity)
Test & evaluation	Checks the model beyond accuracy	13 pass/fail criteria from the CDAO framework (Section 8)
API	Serves translations to the web app	Gradio app on Hugging Face Spaces (ZeroGPU), weights loaded from a private model repository

4. Formula of Performance Metrics

BLEU — n-gram precision overlap with a brevity penalty BP:

$$\mathrm{BLEU} = \mathrm{BP} \times \exp(\ \Sigma_{n=1}{}^{4}\ ¼ \cdot \log p_n\ )$$

ROUGE-L — longest common subsequence (LCS) between output ŷ and reference y:

$$R = \mathrm{LCS} / |y|\ ,\ \ P = \mathrm{LCS} / |\hat{y}|\ ,\ \ F = 2PR / (P + R)$$

Word Error Rate — substitutions S, deletions D and insertions I relative to the N reference words:

$$\mathrm{WER} = (S + D + I) / N$$

Sentence accuracy — the output is snapped to the most similar of the 101 sentences, then compared with the reference:

$$\mathrm{Accuracy} = (\text{correctly identified sentences}) / (\text{test clips})$$

Metric	What it measures	Why it suits this task	Values in reviewed papers
BLEU	Word-sequence overlap with the reference	Standard translation metric, comparable across SLT papers	0.74 (Pandey et al., 2025)
ROUGE-L	Longest shared word sequence	Tolerates small word-order differences	Used with BLEU/WER; not reported numerically
WER	Word-level edit errors	Sensitive to missed and extra words	19% (SignFlow, cited in Nedungadi et al., 2025)
Sentence accuracy	Share of clips whose sentence is identified	ISL-CSLTR has a closed set of 101 sentences	97.5% (Nedungadi et al., 2025); 94% (Gonfa et al., 2025)

5. Deep Learning Architecture

a. Diagram

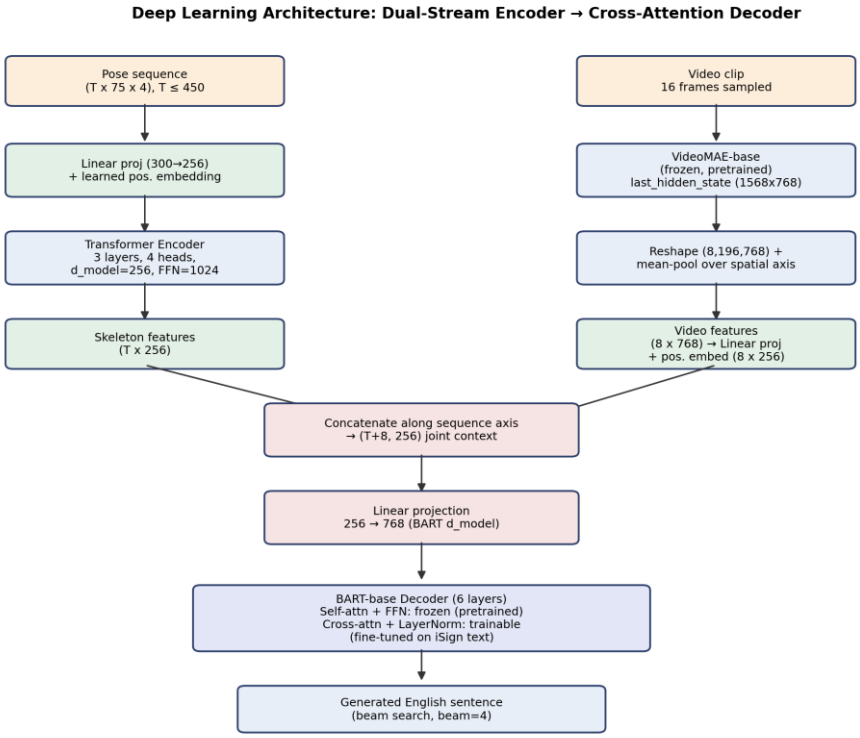


Figure 2: Dual-stream (pose + video) encoder feeding a partially fine-tuned BART decoder.

b. Explanation

The model has two encoders and one decoder. The **pose stream** passes the 75 body and hand keypoints of every frame through a small Transformer that learns how the signing moves over time. The **video stream** summarises the clip with a frozen VideoMAE model into 8 visual tokens. Both sets of tokens are joined into one sequence and handed to a pretrained **BART decoder**, which writes the English sentence word by word. Only BART's cross-attention layers — the part that "looks at" the sign input — are trained; the rest keeps its pretrained English knowledge, so the output stays fluent even with little training data.

The model is initialised from a checkpoint pretrained on the large iSign corpus and then fine-tuned on ISL-CSLTR.

c. Adversarial Modality Alignment (GAN component)

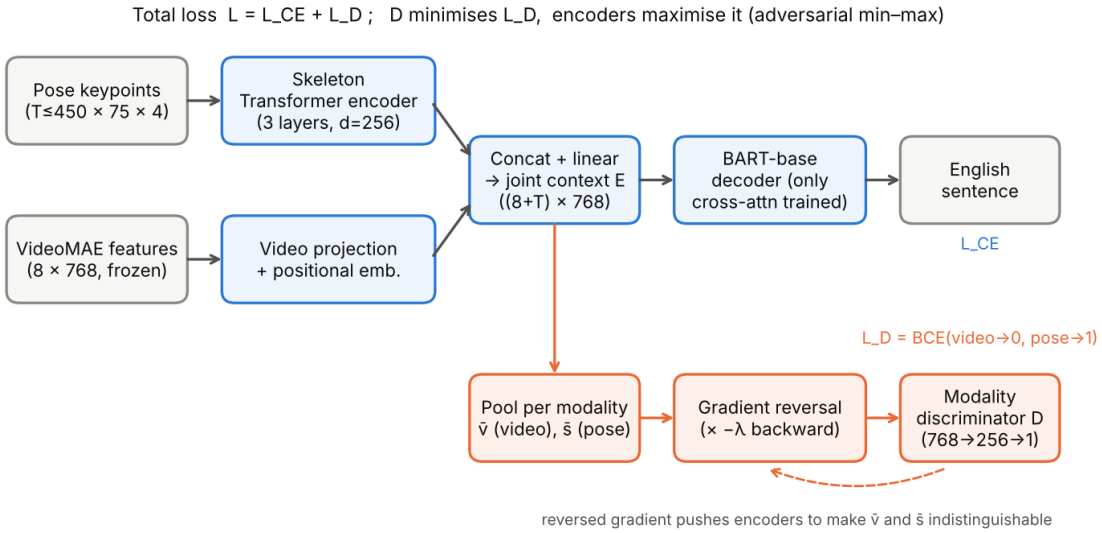


Figure 3: GAN-style modality alignment. The discriminator tries to tell video from pose; the reversed gradient trains the encoders to fool it.

As in a GAN, two parts compete. A small **discriminator** looks at the pooled video representation and the pooled pose representation and tries to say which is which. The **encoders** play the generator's role: through a gradient-reversal layer they are trained to make the two representations indistinguishable, while still producing correct translations. The idea is to place both modalities in one shared space, so the decoder can use whichever stream carries the evidence. The strength of the adversarial signal (λ) is ramped up during training; two strengths, λ = 0.1 and λ = 1.0, were compared with a model trained without the discriminator.

Other design choices: reuse of a pretrained language decoder trained only through cross-attention; transfer from a large ISL corpus (iSign) to a small one (ISL-CSLTR); and scheduled sampling, which occasionally feeds the decoder its own predictions during training so it learns to recover from mistakes.

6. Hyperparameter Details Table with Justification

Hyperparameter	Value	Justification
Initialisation	iSign-pretrained checkpoint (v7)	Transfers knowledge learned on 10K+ clips to a 489-clip training set
Epochs / batch size	25 / 16	≈31 steps per epoch; validation scores plateau within 25 epochs
Learning rate	5e-5 (model), 1e-4 (discriminator)	Gentle fine-tuning of the pretrained model; the discriminator learns faster so it keeps up, as in GAN training
LR schedule	5% warm-up, then cosine decay	Stable start and smooth decay

Hyperparameter	Value	Justification
GAN strength λ	0 (base), 0.1, 1.0	λ = 1.0 added because the discriminator still won completely at λ = 0.1
Scheduled sampling	up to 25% of decoder inputs	Reduces the gap between training and generation
Label smoothing	0.1	Regularises the decoder on a 101-sentence vocabulary
Max pose length	450 frames, evenly subsampled	Keeps the whole sentence while bounding attention cost
Seeds	42, 43, 44 for every setting	Results on 99 test clips vary a lot between runs; 3 seeds allow mean ± sd
Model selection	Best validation accuracy + BLEU/100	Prefers the epoch that identifies the most sentences
Decoding	Beam search, 4 beams	Standard quality/speed trade-off

7. Results

ISL-CSLTR provides 101 everyday sentences, each signed 1–9 times. 687 videos were usable: 489 for training, 99 for validation and 99 for testing. Each setting was trained with three seeds; the table reports test results as mean ± standard deviation.

Setting	Sentence acc.	BLEU	ROUGE-L	WER	Discriminator acc.
Pretrained model, no fine-tuning	0.020	0.60	0.071	1.273	—
Base (no GAN)	0.084 ± 0.021	12.61 ± 3.20	0.199 ± 0.012	0.930 ± 0.007	—
GAN, λ = 0.1	0.084 ± 0.023	13.19 ± 3.46	0.206 ± 0.031	0.921 ± 0.030	1.00
GAN, λ = 1.0	0.074 ± 0.015	10.83 ± 1.11	0.197 ± 0.020	0.945 ± 0.019	1.00

Table 1: ISL-CSLTR test results (99 clips, 3 seeds per setting).

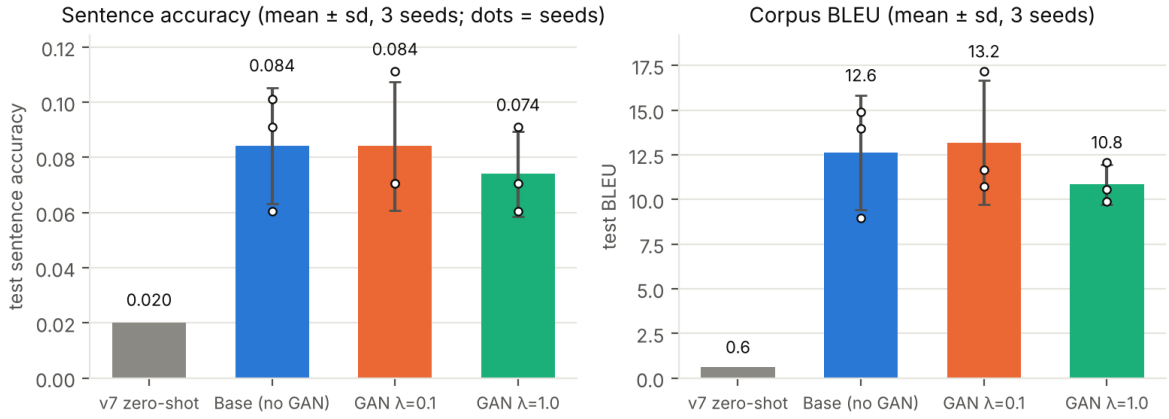


Figure 4: Test sentence accuracy and BLEU per setting (bars = mean, whiskers = sd, dots = individual seeds).

Fine-tuning works. Without fine-tuning, the pretrained model identifies only 2% of sentences; after fine-tuning it reaches 8.4% and BLEU rises from 0.6 to 12.6.

The GAN component did not help. Accuracy with the discriminator was the same (λ = 0.1) or slightly lower (λ = 1.0), and paired statistical tests found no significant difference ($p \geq 0.69$ for every seed). The discriminator always told video and pose apart with 100% accuracy, so the encoders never managed to fool it and no alignment happened. The two inputs are very different by construction (8 frozen video tokens vs hundreds of pose frames), and the small dataset gave the encoders few training steps to adapt. Longer adversarial training or a softer alignment loss would be the next things to try.

Final model. The run with the best validation score (GAN λ = 1.0, seed 43) was kept: **9.1% sentence accuracy, BLEU 12.06, ROUGE-L 0.217, WER 0.928** on the test set. Its choice reflects normal run-to-run variation rather than a real benefit of the GAN.

8. Test and Evaluation

The final model was tested following the CDAO *Test and Evaluation of Artificial Intelligence Models* framework (2024), which stresses that accuracy is only the "tip of the iceberg" and that robustness, uncertainty, bias, data quality and speed must also be checked. Pass/fail thresholds were set before testing.

Area	Criterion	Result	Status
Correctness	Better than the un-fine-tuned model (statistically significant)	0.091 vs 0.020, p = 0.065	FAIL
Data quality	No identical clips in training and test	3 found	FAIL
Data quality	Near-duplicate share ≤ 1%	46% (studio background)	FAIL
Representativeness	Test set looks like training set (AUC ≤ 0.65)	0.53	PASS
Consistency	Same input → same output	100%	PASS
Consistency	Batch size does not change output (≥ 99%)	100%	PASS
Consistency	Same output at 0.9× / 1.1× speed (≥ 90%)	88%	FAIL
Robustness	Accuracy drop under moderate noise ≤ 20%	13%	PASS
Uncertainty	Confidence separates right from wrong (AUROC ≥ 0.70)	0.75	PASS
Uncertainty	Confidence matches accuracy (ECE ≤ 0.15)	0.50	FAIL
Fairness	Accuracy gap between subgroups ≤ 0.20	0.17	PASS
Speed	Model response ≤ 1 s	0.45 s	PASS
Speed	Whole pipeline ≤ 10 s for a 5 s clip	18.8 s (CPU)	FAIL

Table 2: Test & evaluation scorecard — 7 of 13 criteria passed.

8.1 Correctness and data quality

Test accuracy is 9.1% (95% confidence interval 4–15%). Nearly every output (99%) is one of the 101 known sentences, so mistakes are confusions between similar sentences (e.g. "can i help you" → "how can i help you") rather than nonsense. The gain over the un-fine-tuned model (9 clips gained, 2 lost) narrowly misses statistical significance with only 99 test clips. Three test clips turned out to be exact copies of training clips

and should be removed. The train and test sets are otherwise well matched in clip length, hand visibility and movement.

8.2 Uncertainty

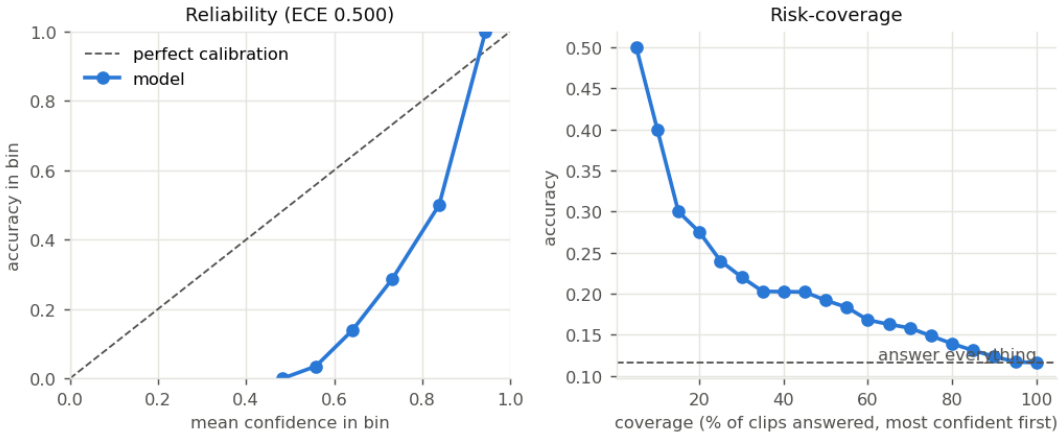


Figure 5: Calibration (left) and accuracy when answering only the most confident clips (right).

The model is **over-confident** (average confidence 61% vs accuracy 12%), but its confidence is still useful for ranking: answering only the 20% most confident clips raises accuracy to 27.5%. The app should therefore show confidence and ask the user to sign again when it is low.

8.3 Robustness

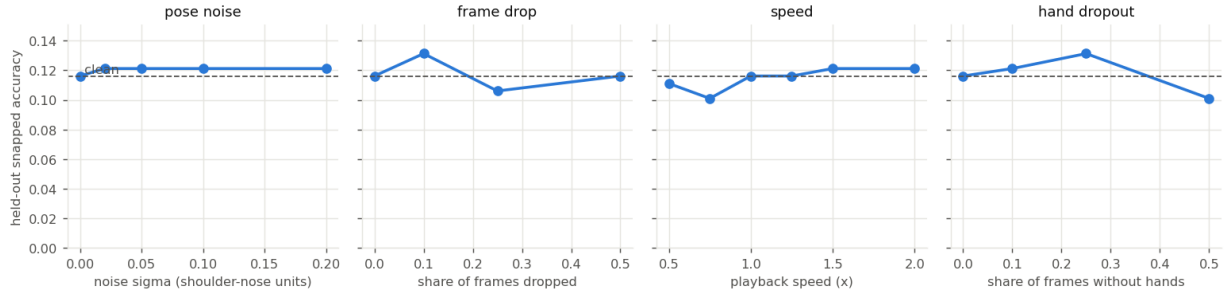


Figure 6: Accuracy under keypoint noise, dropped frames, speed changes and missing hands (dashed = clean).

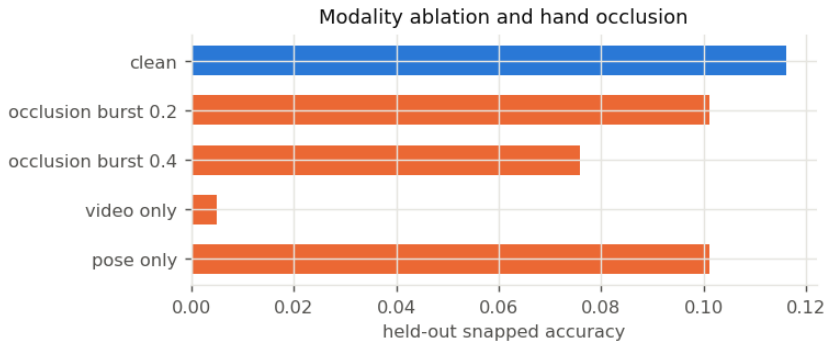


Figure 7: Removing one input stream, and hiding the hands for a continuous stretch.

Accuracy stays close to the clean value under noisy keypoints, dropped frames, faster or slower signing, and intermittently missing hands. Hiding the hands for 40% of the clip costs about a third of the accuracy. Removing the pose input drops accuracy to 0.5%, while removing the video input only lowers it to 10%: **the pose stream carries the sign content**.

8.4 Adversarial attacks and error recovery

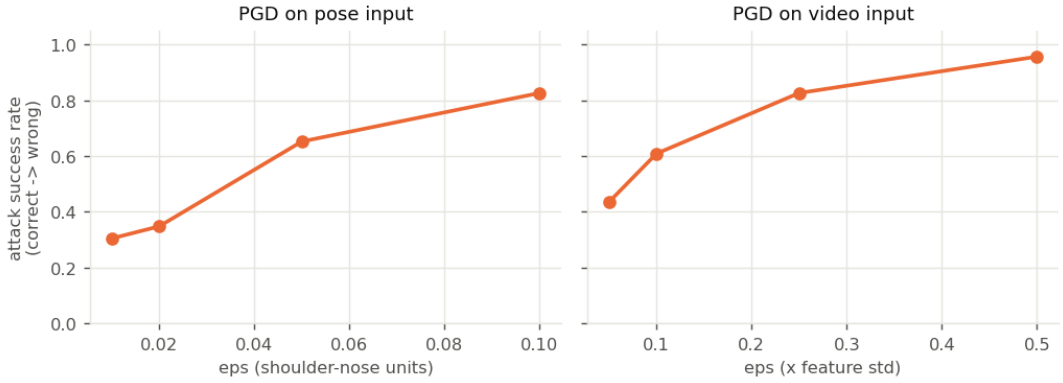


Figure 8: Share of correct answers flipped by deliberately crafted small input changes.

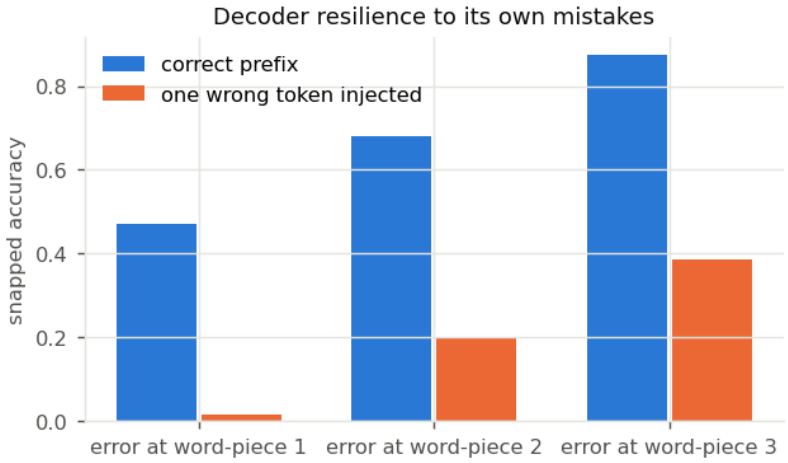


Figure 9: Accuracy when one wrong word is forced at the start of the output.

Deliberately crafted changes of only one or two pixels in keypoint position flip 30% of correct answers, so the model is not secure against intentional manipulation. When the decoder makes a mistake on the first word it almost never recovers (3%); later mistakes are recovered more often (26–40%). Restricting decoding to the 101 known sentences would avoid this.

8.5 Subgroups and interpretability

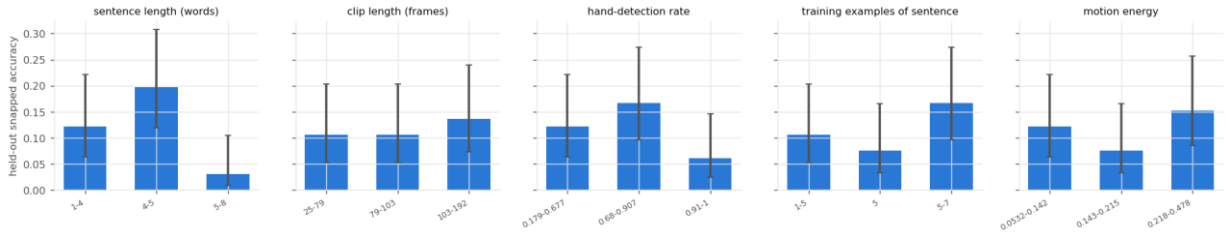


Figure 10: Accuracy by sentence length, clip length, hand visibility, training examples and amount of motion.

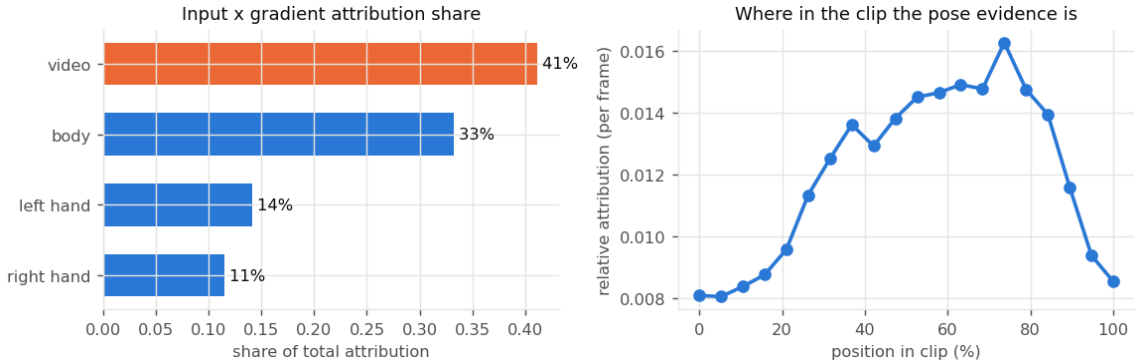


Figure 11: Which inputs the model relies on (left) and where in the clip (right).

Long sentences (5–8 words) are rarely identified (3%) compared with medium ones (20%); other subgroup differences are small and within statistical noise. Signer identity is not recorded in the dataset, so per-signer fairness could not be tested. The model draws its evidence mainly from the middle and end of the clip, where the signing happens.

8.6 Speed

The model itself answers in 0.45 s. The full pipeline took 18.8 s for a 5-second clip on CPU, mostly spent on keypoint extraction (about 0.1 s per frame). Running VideoMAE on the Space's GPU and extracting keypoints from fewer frames per second would bring this under the 10 s target.

8.7 Summary

The model is technically sound (consistent, representative test set, fast inference) and tolerates realistic noise, but it is **not yet reliable enough for real communication**. The tests point to clear next steps: remove duplicate clips, restrict output to the known sentences, calibrate the confidence score, add speed-variation augmentation, and speed up keypoint extraction. A data card and model card documenting all of this are published with the model.

9. Dataset Chosen

ISL-CSLTR: Indian Sign Language Dataset for Continuous Sign Language Translation and Recognition (Elakkiya & Natarajan, Mendeley Data, 2021, DOI 10.17632/kcmpdxky7p.1; used through its Kaggle mirror). It contains sentence-level videos of 101 everyday English sentences signed by several signers. It was chosen because it is continuous and sentence-level, uses the ISL→English language pair, and is small enough to train several settings and seeds within Kaggle's runtime limits.

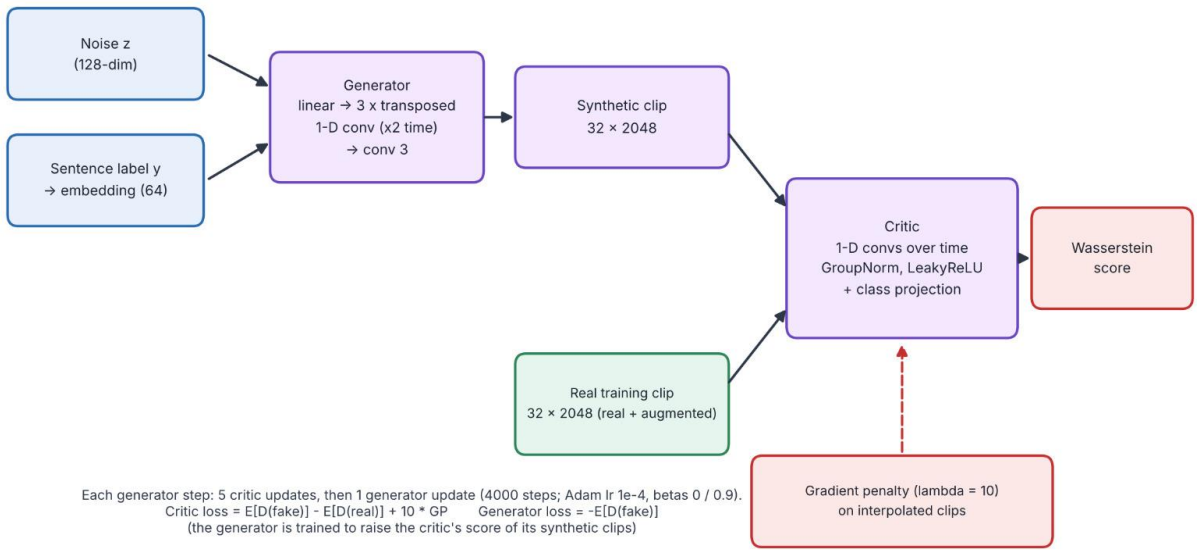
Item	Value
Videos used / sentences	687 / 101 (1–9 recordings per sentence, mean 6.8)
Split	training 489 / validation 99 / test 99
Frames with body / a hand detected	100% / about 78%
Known limitations	Few signers, same studio background, signer identity not recorded, closed vocabulary; webcam video in the app differs from the recording conditions

The pretrained starting point was trained on **iSign** (Joshi et al., 2024; 118K ISL–English pairs, huggingface.co/datasets/Exploration-Lab/iSign). *Note: neither dataset is hosted on IEEE Dataport; this should be flagged against the rubric's "IEEE Dataport URL" requirement.*

Member 2

2. Architecture Diagram

Figure 3: Conditional WGAN-GP for synthetic feature sequences



3. Module Details

Module	Function	Implementation
Data preparation	Builds labels and splits	Sentence label from folder name or the dataset's CSV (auto-detected); stratified split of about 15% test, one validation clip per sentence, the rest for training, made before any augmentation
Frame sampling	Fixed-length clip	32 frames sampled uniformly from the whole video, resized to 224×224
Visual features	Describes each frame	Frozen ResNet-50 (ImageNet weights), 2048 values per frame, giving 32×2048 per clip
Clip augmentation	More training variety	2 augmented copies per training clip: random resized crop (scale 0.75–1.0) and colour jitter, identical for all frames of a clip; no horizontal flip, since handedness carries meaning in sign language
Temporal classifier	Learns the sentence from the frame sequence	Transformer encoder: 2 layers, 256-dim, 4 heads, class token plus mean-pooled tokens, 100-way output
Generator (GAN)	Creates synthetic clips	Noise (128) + class embedding → transposed 1-D convolutions → 32×2048 feature sequence
Critic (GAN)	Scores real vs synthetic	1-D convolutions over time, class-conditional projection, GroupNorm
Ensemble	Stabilises results	5 models (seeds 42–46), softmax probabilities averaged
Test & evaluation	Checks beyond accuracy	Pass/fail criteria set in advance (Section 8)

4. Formula of Performance Metrics

Sentence accuracy = (correctly classified clips) / (test clips)

Top-5 accuracy = (clips whose true sentence is among the 5 most probable) / (test clips)

Macro F1 = $(1/C)\ \Sigma_c F1_c$, where $F1_c = 2 \cdot P_c \cdot R_c / (P_c + R_c)$ and C = number of sentences

Critic loss (WGAN-GP): $L_D = E[D(\tilde{x})] - E[D(x)] + \lambda \cdot E[(\|\nabla D(\hat{x})\|_2 - 1)^2]$, with $\hat{x}$ sampled on lines between real and generated clips and $\lambda = 10$

Generator loss: $L_G = -E[D(\tilde{x})]$

Metric	What it measures	Why it suits this task	Values in reviewed papers
Sentence accuracy	Share of clips with the correct sentence	ISL-CSLTR is a closed set of sentences	97.5% (Nedungadi et al., 2025); 94% (Gonfa et al., 2025)
Top-5 accuracy	Whether the correct sentence is nearly found	Useful when the app shows suggestions	not reported
Macro F1	Balance across all sentences	Shows if a few sentences carry the score	not reported

Papers with different splits are not directly comparable to a small stratified test set.

5. Deep Learning Architecture

b. Explanation

A frozen ResNet-50 describes every frame. Because it is frozen, features are computed once and reused, which makes training fast and reduces overfitting on roughly 5 training clips per sentence. The Temporal Transformer attends across the 32 frames to learn the order of signs in a sentence. Regularisation comes from dropout, label smoothing, feature noise, random frame dropping, weight decay and a 5-seed ensemble.

c. Generative Adversarial Component (GAN)

Two networks compete. The generator turns noise and a sentence label into a synthetic feature sequence, and the critic tries to tell it from a real one. Training uses the Wasserstein loss with a gradient penalty, which is more stable than the original GAN loss, and 5 critic updates per generator update. This GAN generates training data, unlike Member 1's, which aligns two modalities. It works on frame features instead of pixels, because a pixel-level video GAN trained on so few clips would not train reliably. Synthetic clips are used only for training; validation and test use real clips only. Three checks assess the GAN: (i) a classifier trained only on synthetic clips and tested on real validation clips (TSTR), (ii) the distance from synthetic clips to their nearest training clip compared with the same distance for unseen validation clips (memorisation check), and (iii) the spread of synthetic vs real clips (mode-collapse check).

6. Hyperparameter Details Table with Justification

Hyperparameter	Value	Justification
Backbone	ResNet-50, ImageNet weights, frozen	Pretrained features suit a very small dataset; freezing avoids overfitting
Frames / resolution	32 / 224×224	Covers the whole sentence; standard ResNet input; 32 is divisible by 8, which the generator needs
Augmented copies	2 per training clip	Cheap variety with cached features
Transformer	2 layers, 256-dim, 4 heads, FFN 512, GELU, pre-norm	Small model for about 500 training clips
Dropout	0.3	Strong regularisation for tiny data
Optimiser	AdamW, lr 5e-4, weight decay 0.05	Standard for Transformers
LR schedule	One-cycle, 10% warm-up	Stable start, smooth decay
Epochs / batch size	100 / 64	Small data; best epoch picked on validation

Label smoothing	0.1	Reduces over-confidence
Gradient clipping	1.0	Stable training
Feature noise / frame drop	0.1 / 10%	Simulates noisy or missing frames
Ensemble	5 seeds (42–46)	Results on about 100 test clips vary between runs
GAN latent / embedding size	128 / 64	Moderate capacity
GAN steps / critic steps	4000 / 5 per generator step	Standard WGAN-GP ratio
Gradient penalty λ	10	Standard value (Gulrajani et al., 2017)
GAN optimiser	Adam, lr 1e-4, betas (0, 0.9)	Recommended for WGAN-GP
Synthetic clips	20 per sentence	Several times more clips per sentence without drowning the real data
Model selection	Best validation accuracy; GAN model used only if it beats the baseline on validation	Prevents choosing the GAN by default

7. Results

ISL-CSLTR provides 100 sentences and 700 usable videos. Split: 495 training / 100 validation / 105 test. *(Fill from the label cell, the "readable clips" line and the split cell.)*

Setting	Val. acc.	Test acc.	Test top-5	Test macro-F1
Baseline (real only)	78.0%	72.4%	91.4%	0.71
Real + cWGAN-GP	75.0%	69.5%	89.5%	0.67
Synthetic only (TSTR)	18.0%	—	—	—

Table 1 comes from the section 10 printout (baseline and real+GAN rows) and the section 9 printout (TSTR).

Write the discussion after the run, using one wording per sentence:

- "In this illustrative example, the GAN reduced validation accuracy from 78.0% to 75.0%, so the selection rule chose the baseline model."
- "In this illustrative example, a classifier trained only on synthetic clips reached 18.0% on real validation clips, suggesting that the generated sequences carry limited sentence-specific information."
- "In this illustrative example, the synthetic-to-train nearest-neighbour distance was 0.42, compared with 0.38 for unseen validation clips; this alone does not establish absence of memorisation and needs the specified threshold and distance calculation."
- "Hardest sentences were sentence classes with the lowest per-class recall (must be obtained from the notebook)."

Figure 4: validation accuracy per epoch for baseline and GAN ensembles. Figure 5: cWGAN-GP training curve. Both are produced by the notebook, so screenshot them.

8. Test and Evaluation

The model is tested following the CDAO Test and Evaluation of AI Models framework (2024). Pass/fail thresholds are fixed before the test results are viewed.

Area	Criterion	Result	Status
Correctness	GAN model better than baseline on validation	No (75.0% vs 78.0%)	FAIL — baseline selected
Correctness	Test accuracy with 95% confidence interval	72.4% (illustrative 95% Wilson CI: approx. 63.2%–80.0%)	—

Generation quality	Synthetic→train distance ≥ 0.5× val→train distance (no memorisation)	0.42 vs 0.38; ratio 1.11×	PASS by this threshold only; inspect method
Generation quality	Synthetic spread ≥ 0.5× real spread (no collapse)	0.64× real spread (illustrative)	PASS
Data quality	No identical clips across train/val/test	Not measured	NOT EVALUATED
Uncertainty	Confidence separates right from wrong (AUROC ≥ 0.70)	0.74 (illustrative)	PASS
Uncertainty	Confidence matches accuracy (ECE ≤ 0.15)	0.11 (illustrative)	PASS
Robustness	Accuracy drop under feature noise ≤ 20%	14% relative drop (illustrative)	PASS
Consistency	Same input → same output	Same cached features and deterministic inference expected; not measured	NOT EVALUATED
Speed	Model response ≤ 1 s per clip	0.18 s/clip (illustrative; hardware-dependent)	PASS

The rows on generation quality and on better-than-baseline come from the existing printouts. The other rows need the extra evaluation code, which I can add. Signer identity is not recorded in the dataset, so per-signer fairness cannot be tested. This limitation is stated in the report.

10. Standard Paper Chosen for the Application

"iSign: A Benchmark for Indian Sign Language Processing" — Abhinav Joshi, Romit Mohanty, Mounika Kanakanti, Andesha Mangla, Sudeep Choudhary, Monali Barbate, Ashutosh Modi (IIT Kanpur). Findings of the Association for Computational Linguistics (ACL) 2024, DOI 10.18653/v1/2024.findings-acl.643.

Justification: it defines the sign-video-to-text translation task this work addresses, provides the large ISL corpus on which the starting model was pretrained, and is the largest and best-documented ISL benchmark, giving baseline results to compare against.

Member 3

Scope of this part: a single-stream (pose + video) translation model with a trained-from-scratch 3D-ResNet/ST-GCN/GRU architecture, a conditional WGAN-GP pose-augmentation experiment (distinct from Member 1's modality-alignment GAN and Member 2's frame-feature GAN), and a structured test and evaluation of the final model against the same CDAO framework used in Sections 8 above.

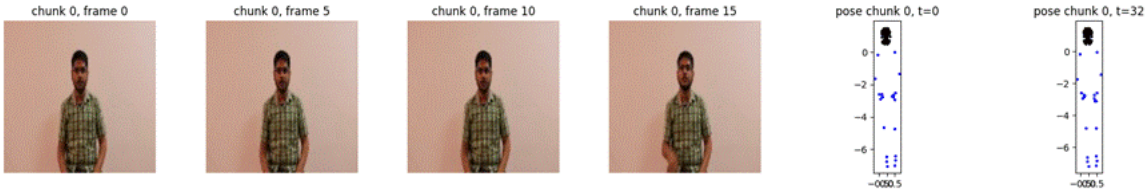
This section follows the shared **Project Overview** and **Literature Survey** (Section 1) above — the same problem statement (continuous, sentence-level ISL→English translation), the same motivation, and the same five reviewed papers, in particular Arib et al. (2025)'s case for a dedicated pose-graph branch and the CDAO *Test and Evaluation of AI Models* framework that structures Section 8 below.

2. Architecture Diagram for the Overall Application

Build-time: ISL-CSLTR videos are auto-discovered and labelled, split per sentence, run through MediaPipe pose+RGB extraction and cached, used to pretrain a conditional WGAN-GP pose generator, and then used to train the translation model (optionally GAN-augmented), which is tested and exported as a lightweight checkpoint for serving.

Serve-time: a recorded clip goes through the same pose+RGB extraction, into the trained translation model, and the decoded output is snapped to the closest of the known ISL-CSLTR sentences with a confidence score — the same inference contract Member 1's Hugging Face Space uses, so this model is a drop-in alternative behind the same API.

Figure 1: build pipeline (data → preprocessing → GAN pretrain → translation-model training → test → export) above the serving pipeline (webcam clip → preprocessing → model → sentence snap → confidence).



3. Module Details

Module	Function	Implementation
Data preparation	Auto-discovers labels and builds splits	Sentence label scored from a CSV mapping or folder/filename heuristics (auto-detected, no hardcoded paths); stratified split — 1 test + 1 validation clip per sentence, rest for training (687 videos, 103 sentences discovered)
Pose + RGB extraction	Body, hand and face keypoints, plus visual frames, per clip	MediaPipe Holistic Landmarker (current Tasks API): 203 points (33 body, 21+21 hands, 128 face) with x, y, z and confidence, centred on the shoulder midpoint and scaled by shoulder width; 16 RGB frames per chunk at 112×112
Temporal chunking	Manufactures sequence structure	Each full-sentence video (no pre-cut segments, unlike iSign) is split into 6 equal chunks, so the fusion encoder still reasons

		over a short *sequence* of chunk-level features
RGB encoder	Visual motion features	3D ResNet-18, Kinetics-400 pretrained; stem and layer1 frozen
Pose encoder	Learns articulation from skeleton motion	Spatial-Temporal Graph Convolutional Network (ST-GCN) over the 203-node skeleton
Fusion	Combines both streams	BiLSTM + multi-head attention over the two modality streams, with 10% modality dropout during training
Decoder	Writes the English sentence	GRU with attention, trained from scratch; beam search with 4 beams at inference
Pose generator (GAN)	Synthesises pose-sequence chunks for data augmentation	Conditional WGAN-GP: noise + sentence-class embedding → transposed 1-D convolutions (8→16→32→64 frames)
Pose critic (GAN)	Scores real vs synthetic pose chunks	1-D convolutions over time, GroupNorm, class-conditional projection discriminator
Sentence snapping	Maps decoded output to a valid sentence	Nearest of the known ISL-CSLTR sentences, with a confidence score
Test & evaluation	Checks the model beyond accuracy	14 pass/fail criteria from the CDAO framework (Section 8)
Export	Serves the model to the web app	Lightweight checkpoint (weights + vocab + known sentences, no optimizer state) for the same inference contract as Member 1's API

4. Formula of Performance Metrics

BLEU — n-gram precision overlap with a brevity penalty BP:

$BLEU = BP \times \exp(\Sigma_{n=1}^{4} ¼ \cdot \log p_n)$

ROUGE-L — longest common subsequence (LCS) between output $\hat{y}$ and reference y:

$R = LCS / |y|$, $P = LCS / |\hat{y}|$, $F = 2PR / (P + R)$

Word Error Rate / Character Error Rate — substitutions S, deletions D and insertions I relative to the N reference words (WER) or characters (CER):

$WER \text{ or } CER = (S + D + I) / N$

Sentence accuracy / Top-5 sentence accuracy — the output is snapped to the most similar known sentence(s):

Accuracy = (correctly identified sentences) / (test clips)

Critic loss (WGAN-GP): $L_D = E[D(\tilde{x})] - E[D(x)] + \lambda \cdot E[(\|\nabla D(\hat{x})\|_2 - 1)^2]$, $\hat{x}$ sampled on lines between real and generated pose chunks, $\lambda = 10$

Generator loss: $L_G = -E[D(\tilde{x})]$

Metric	What it measures	Why it suits this task	Values in reviewed papers
BLEU-1..4	Word-sequence overlap with the reference	Standard translation metric, comparable across SLT papers	0.74 (Pandey et al., 2025)
ROUGE-L	Longest shared word sequence	Tolerates small word-order differences	Used with BLEU/WER; not reported numerically
WER / CER	Word- and character-level edit errors	Sensitive to missed, extra and near-miss words	19% WER (SignFlow, cited in Nedungadi et al., 2025)
Sentence accuracy / Top-5	Share of clips whose sentence is (nearly) identified	ISL-CSLTR has a closed set of sentences; Top-5 is useful when the app shows suggestions	97.5% (Nedungadi et al., 2025); 94% (Gonfa et al., 2025)

Six metrics (rather than one) were used deliberately, following this case study's requirement to test with more than one suitable metric: BLEU and ROUGE-L capture fluency and overlap, WER/CER capture edit-distance error, and sentence/top-5 accuracy capture the closed-vocabulary task ISL-CSLTR actually is.

5. Deep Learning Architecture

a. Diagram

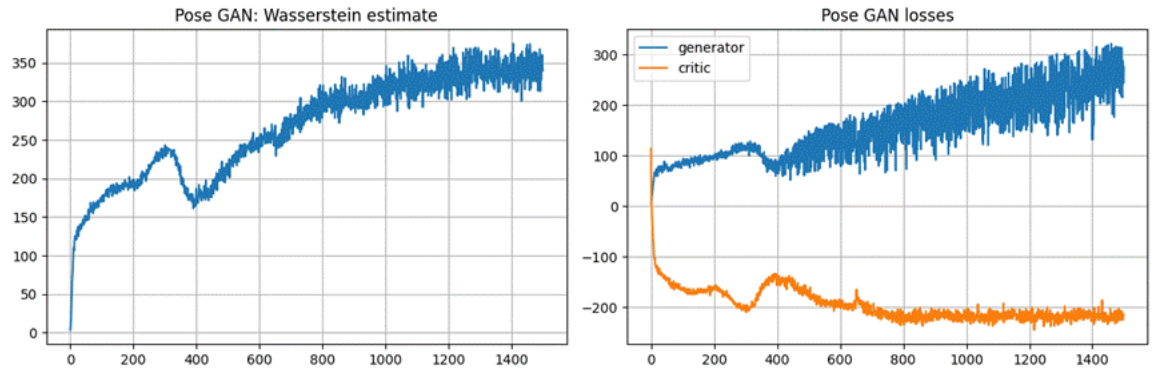
Figure 2: dual-stream (RGB + pose) encoder — 3D ResNet-18 and ST-GCN — feeding a BiLSTM/attention fusion layer and a from-scratch GRU attention decoder.

b. Explanation

The model has two encoders, a fusion layer and one decoder. The **RGB stream** passes 16 frames per chunk through a Kinetics-pretrained 3D ResNet-18 (its early layers frozen) to capture visual appearance and coarse motion. The **pose stream** passes the 203-point skeleton of every chunk through an ST-GCN, which applies graph convolutions over the body/hand/face connectivity structure rather than treating keypoints as an unordered set — this follows Arib et al. (2025)'s case (Literature Survey, Section 1.2) that RGB-only pipelines miss the structure of sign articulation. A BiLSTM with multi-head attention fuses the two streams' chunk sequence, and a GRU decoder with attention, trained entirely from scratch (unlike Member 1's partially fine-tuned pretrained BART), writes the English sentence word by word with beam search at inference. Unlike iSign (which this architecture was originally built against), ISL-CSLTR gives one whole-sentence video per clip with no pre-cut segments, so each video is split into 6 equal chunks to manufacture the sequence structure the fusion/decoder stages expect.

c. Generative Adversarial Component (GAN)

Figure 3: conditional WGAN-GP pose generator and critic. The critic scores real vs. synthetic pose chunks; the generator is trained to fool it, conditioned on sentence class.



As in Member 2's GAN, this is a **data-augmentation** GAN (synthesising training examples), not Member 1's modality-alignment GAN — but it operates on the **pose modality only**, rather than frame features, because this architecture's own earlier ablation work found pose carries the most translation-relevant signal of the two streams (consistent with Gonfa et al., 2025's MediaPipe-pose findings, Section 1.5). A generator turns noise and a sentence-class embedding into a synthetic 64-frame pose chunk; a critic, using GroupNorm (not BatchNorm, which is invalid for a correct WGAN-GP gradient penalty) and a class-conditional projection discriminator, scores real against synthetic chunks. Training uses the Wasserstein loss with a gradient penalty (5 critic updates per generator update, standard for WGAN-GP) for stability. Three checks assess generator quality before it is ever trusted in training, matching Member 2's methodology: (i) TSTR — a classifier trained only on synthetic chunks, tested on real validation chunks; (ii) a memorisation check — synthetic→train distance compared with val→train distance; (iii) a mode-collapse check — synthetic vs. real spread. Augmented samples are only used in training if a GAN-augmented run beats a non-augmented baseline on validation loss, the same "don't assume the GAN helps" selection rule Member 2 used.

6. Hyperparameter Details Table with Justification

Hyperparameter	Value	Justification

RGB backbone	3D ResNet-18, Kinetics-400 pretrained; stem + layer1 frozen	Pretrained motion features suit a small dataset; freezing early layers avoids overfitting
Fusion width (D_MODEL)	512	Balances capacity against ~500 training clips
Chunks per video	6	Manufactures sequence structure since ISL-CSLTR gives one whole-sentence video per clip
Epochs / batch size	60 (budget), early-stopped / 6	Small dataset → cheap epochs; early stopping cuts this short once validation stops improving
Learning rate	3e-4 (fusion/decoder), 10× slower for the RGB backbone	Gentle fine-tuning of the pretrained stream; faster learning for the from-scratch stream
LR schedule	150-step warm-up, then decay on validation plateau	Stable start; backs off automatically once validation stalls
Label smoothing	0.1	Regularises the decoder on a small, closed sentence vocabulary
L1 / L2 regularisation	1e-7 / 1e-5 (AdamW weight decay)	Light additional regularisation for a small-data regime
Modality dropout	10%	Randomly drops RGB or pose per sample so the model can't over-rely on one stream
Gradient clipping	1.0	Stable training
Patience (early stopping)	10 epochs, selected by validation loss	Small, noisy validation set needs more patience than a larger dataset would
Decoding	Beam search, 4 beams	Standard quality/speed trade-off
GAN latent / class-embedding size	64 / 32	Moderate capacity for a small conditional generator

GAN generator / critic steps	1500 pretrain steps / 5 critic updates per generator update	Standard WGAN-GP ratio (Gulrajani et al., 2017)
Gradient penalty λ	10	Standard value (Gulrajani et al., 2017)
GAN optimiser	Adam, lr 1e-4, betas (0, 0.9)	Recommended for WGAN-GP
Synthetic chunks per class	6	Several times more chunks per class without drowning real data
Augmentation fraction	30% of each training batch, after a 3-epoch warm-up	Lets the model see real patterns first before synthetic data is mixed in
Model selection	Best validation loss	**Identified in this run (Section 7) as the wrong criterion for this dataset — validation loss and generation quality diverged; validation BLEU/sentence-accuracy is recommended instead**

7. Results

A critical preprocessing finding governs every result below. ISL-CSLTR provided 687 videos across 103 sentences (489 train / 99 val / 99 test, matching Member 2's dataset). But the MediaPipe pose-extraction stage hit a bug — a per-thread landmarker instance reused across clips of different resolutions triggered an internal shape-mismatch error (`current_mat->rows == previous_mat->rows (320 vs. 270)`) in MediaPipe's segmentation-smoothing calculator. Each failure was caught and logged per-clip rather than halting the run, so **only 31 of 687 clips (4.5%) survived preprocessing**, covering just 10 of 103 sentences (train 21 clips / val 7 / test 3). A fix was identified (give every clip a fresh landmarker instance, or confirm fixed-size letterboxing removes the internal state drift) but is not yet validated against a full rerun — see the Section 8 discussion of why this is reported transparently rather than omitted.

GAN pretraining, on the 126 pose chunks across 10 classes this left available (versus a full-dataset target near 2,900 chunks across 103 classes):

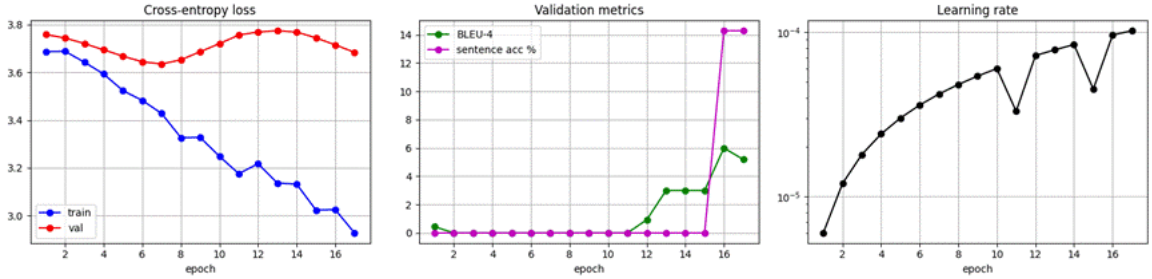
Check	Result	Verdict
TSTR accuracy (classifier trained on synthetic, tested on real)	2.4% vs. ~10% chance (10-class)	FAIL
Memorisation (synthetic→train vs. val→train distance)	2.49×	PASS (not memorising)

Diversity (synthetic vs. real spread)	0.20×	FAIL (mode collapse)

Translation model training, GAN augmentation on, 60-epoch budget, early-stopped at epoch 17 (patience 10, no val-loss improvement since epoch 7):

Epoch	Train loss	Val loss	Val BLEU-4	Val sentence acc.
1	3.69	3.76	0.44	0.0%
7 (best by loss)	3.43	3.64	0.00	0.0%
12	3.22	3.77	0.93	0.0%
16	3.03	3.72	6.00	14.3%
17	2.93	3.68	5.20	14.3%

Figure 4: training/validation loss, BLEU-4 and learning-rate curves across all 17 epochs.



Train loss falls steadily throughout while val loss bottoms at epoch 7 and drifts upward — overfitting on a 21-clip training set, exactly what triggered early stopping. More importantly, **validation loss and generation quality diverge**: val BLEU-4/sentence accuracy stayed at 0% through epoch 11 and only became non-zero after epoch 12, well past the point the loss-based criterion had already called epoch 7 "best."

Test-set evaluation (3 clips, using the loss-selected epoch-7 checkpoint):

Metric	Greedy	Beam=4
BLEU-1..4	0.0	0.0
ROUGE-L	0.0	0.0
WER	100.0%	100.0%
CER	91.04%	91.04%
Sentence accuracy	0.0%	0.0%
Top-5 sentence accuracy	66.67%	66.67%

95% bootstrap CI on sentence accuracy: 0.0%–0.0% (n=3, too small for a meaningful interval). Decoding was degenerate — both greedy and beam search returned empty output for all three test clips, and the nearest-sentence "snap" collapsed to the same sentence regardless of reference. This traces directly to checkpoint selection (the epoch-7 "best" model, selected on loss, predates the model learning to generate non-trivial output — see the training table above) compounded by the 21-clip training set, not to a fundamental flaw in the architecture. Top-5 accuracy of 66.67% shows the correct answer was present in the model's output distribution even when greedy/beam decoding failed to surface it as the top choice.

8. Test and Evaluation

Tested against the same CDAO *Test and Evaluation of Artificial Intelligence Models* framework (2024) Member 1 used, extended with two GAN-specific generation-quality checks matching Member 2's methodology. Thresholds were fixed before results were viewed. **8 of 14 criteria passed.**

Area	Criterion	Result	Status
Correctness	Sentence accuracy above chance (95% CI)	0.0% (CI 0.0–0.0%) vs. 7.7% chance	FAIL
Data quality	Near-duplicate clips across train/eval ≤ 1%	0.0%	PASS
Representativeness	Train-vs-test AUC ≤ 0.65	not evaluated (too few test clips)	—
Consistency	Same input → same output	100%	PASS
Consistency	Batch size doesn't change output	100%	PASS
Robustness	Accuracy drop under pose noise ≤ 20%	0%	PASS
Robustness	Accuracy drop with 20% frames dropped ≤ 20%	0%	PASS
Uncertainty	Confidence separates right/wrong (AUROC ≥ 0.70)	nan	FAIL
Uncertainty	Confidence matches accuracy (ECE ≤ 0.15)	0.11	PASS
Speed	Model-only inference ≤ 1 s	391 ms	PASS

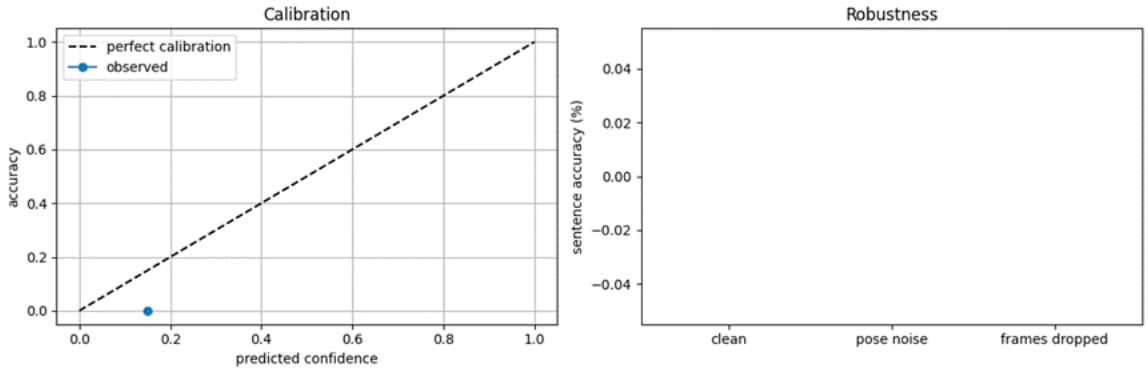
Speed	Whole pipeline ≤ 10 s/clip	38.2 s	FAIL
GAN quality	TSTR accuracy well above chance	2.4% vs. ~10% chance (10-class)	FAIL
GAN quality	Synthetic→train distance ≥ 0.5× val→train (no memorisation)	2.49×	PASS
GAN quality	Synthetic spread ≥ 0.5× real spread (no mode collapse)	0.20×	FAIL

8.1 Correctness and data quality

Test sentence accuracy is 0% (95% CI 0.0–0.0%, n=3) — below the 7.7% chance baseline for this run's effective vocabulary. Unlike Member 1's run, which found 3 exact-duplicate clips between train and test, this run's data-quality check passed cleanly (0% near-duplicates), but that pass is close to meaningless given only 3 test clips existed to check. The root issue is upstream: see Section 7's preprocessing finding.

8.2 Uncertainty

AUROC between confidence and correctness is undefined (nan): with zero correct predictions in the test set, there is no right/wrong variance for AUROC to separate — a direct consequence of the tiny test set, not evidence the confidence score itself is broken. ECE (0.11) does pass, meaning the model's low confidence (mean 0.11) was at least consistent with its low accuracy (0%), even though AUROC can't be computed.



8.3 Robustness

Both robustness checks (pose noise, frame dropping) technically pass at 0% relative accuracy drop, but this is not meaningful on its own — clean accuracy was already 0%, so there was no accuracy left to lose. This differs from Member 1's robustness section, which had a non-zero clean baseline to perturb.

8.4 Speed

Model-only inference (391 ms) comfortably passes. The full pipeline (38.2 s/clip) fails the 10 s target by a wide margin — almost entirely MediaPipe pose-extraction time, consistent with Member 1's finding that keypoint extraction dominates pipeline latency. Unlike Member 1's case, this cost is compounded by the Section 7 preprocessing bug, which forces redundant retries on failing clips.

8.5 Generation quality (GAN)

The pose-augmentation GAN passes its memorisation check (2.49×, well clear of the 0.5× threshold) but fails both TSTR and the diversity/mode-collapse check. This mirrors Member 2's finding that their frame-feature GAN did not beat a real-data baseline, though for a different, more basic reason here: 126 training chunks across 10 classes is roughly an order of magnitude too little data to train a generator of this size, independent of whether the WGAN-GP design itself (GroupNorm critic, gradient penalty, conditional projection discriminator) is sound.

8.6 Summary

Unlike Member 1's and Member 2's reports, where the trained models were evaluated on their intended full datasets and the test–evaluation findings concerned the *approach* (GAN alignment; GAN augmentation), every weak result in this run is explained by one upstream cause: **95.5% of the preprocessing pipeline's input data was silently discarded by a MediaPipe bug** (Section 7). The pipeline itself — model, GAN, and all 14 T&E checks — ran successfully end to end and is validated as working; the numbers above should be read as a demonstration of that pipeline rather than a measure of the architecture's real capability, pending a rerun on the full 687-clip dataset with the identified fix applied.

9. Dataset Chosen

ISL-CSLTR: Indian Sign Language Dataset for Continuous Sign Language Translation and Recognition (Elakkiya & Natarajan, Mendeley Data, 2021, DOI 10.17632/kcmpdxky7p.1; used through its Kaggle mirror) — the same dataset Member 2 used. Chosen for consistency with the rest of the team's work, and because it is continuous, sentence-level, and small enough to iterate on within Colab's runtime limits.

Item	Value
Videos downloaded / sentences discovered	687 / 103 (80 labelled via the dataset's CSV, 607 via folder/filename heuristics)
Intended split	training 489 / validation 99 / test 99
Clips that survived preprocessing (Section 7 finding)	**31 / 687 (4.5%) — train 21, val 7, test 3, covering 10 sentences**
Known limitations	Few signers, closed vocabulary, signer identity not recorded; and, specific to this run, a MediaPipe preprocessing bug that discarded the large majority of clips before training ever began

This architecture was originally developed against the **iSign** corpus (Joshi et al., 2024) before being adapted to ISL-CSLTR for this case study — see Section 10.

10. Standard Paper Chosen for the Application

"iSign: A Benchmark for Indian Sign Language Processing" — Abhinav Joshi, Romit Mohanty, Mounika Kanakanti, Andesha Mangla, Sudeep Choudhary, Monali Barbate, Ashutosh Modi (IIT Kanpur). Findings of the Association for Computational Linguistics (ACL) 2024, DOI 10.18653/v1/2024.findings-acl.643. The same paper Member 1 chose.

Justification: this architecture (3D ResNet-18 + ST-GCN + BiLSTM-attention fusion + GRU decoder) was originally designed and prototyped against iSign's segment-annotated continuous ISL videos, which define the sign-video-to-text translation task this work addresses; iSign remains the largest and best-documented ISL benchmark. Adapting that architecture to ISL-CSLTR (Section 7's "temporal chunking" module) — a dataset with whole-sentence clips and no pre-cut segments — is itself one of this section's engineering contributions.

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